

Pranathi Deepak

Machine Learning Engineer

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Professional Summary

Machine Learning Engineer with 4+ years of experience shipping production ML and Generative AI systems across financial services (KPMG) and applied AI environments. Recently completed a Master's in Computer Science with research focus on Generative AI and RAG systems. Hands-on builder of end-to-end ML pipelines using Python, PyTorch, and AWS, from feature engineering through real-time inference and MLOps. Recent work focuses on LLM and RAG systems with LangChain and LlamaIndex, including a multi-agent learning platform built on Llama 3.3 with adaptive content generation and evaluation. Strong on production deployment, MLOps lifecycle, and stakeholder communication on model governance.

Technical Skills

Programming Languages: Python, SQL, Java

Generative and Agentic AI: AI Agents, Multi-Agent Systems, LLMs, RAG (Retrieval-Augmented Generation), Prompt Engineering, Fine-Tuning, LangChain, LlamaIndex, Vector Databases (FAISS, Pinecone)

Machine Learning: XGBoost, Random Forest, Gradient Boosting, SVM, Logistic Regression, Clustering, Scikit-learn, Hyperparameter Tuning, Bayesian Optimization, Cross-Validation, SHAP, ROC-AUC

Deep Learning and NLP: PyTorch, TensorFlow, Transformers, Sentence-Transformers, CNNs, RNNs, LSTMs, Embeddings, Text Classification, Semantic Search

MLOps and Cloud: AWS (S3, EC2, SageMaker, Lambda, Redshift), Azure ML, GCP Vertex AI, Docker, Kubernetes, Apache Airflow, MLflow, CI/CD, Model Versioning, Drift Detection, Model Monitoring

Data Engineering and Streaming: Pandas, NumPy, Apache Spark, PySpark, Spark Streaming, Kafka, Databricks, Distributed Systems, ETL/ELT Pipelines, Feature Stores, Data Validation

APIs and Deployment: FastAPI, Flask, REST APIs, Batch Inference, Real-Time Inference, Low-Latency Inference, Model Serving

Mathematics and Statistics: Linear Algebra, Probability, Statistical Inference, Hypothesis Testing, Optimization

Work Experience

Machine Learning Engineer – Graduate Research

Aug 2024 – Dec 2025

California State University

Carson, CA

- Designed and deployed a Retrieval-Augmented Generation pipeline in Python using LangChain, LlamaIndex, and FAISS vector databases that grounds LLM responses in domain-specific documents, cutting average retrieval time by 40% on internal test queries.
- Tuned prompts and retrieval strategies (chunking, overlap, system prompts) and fine-tuned sentence-transformers on domain text, iterating with stakeholder feedback to reduce off-topic and unsupported responses.
- Developed embedding-based semantic search using sentence-transformer models indexed in FAISS, surfacing relevant content that simple keyword matching consistently missed.
- Wrapped model inference in containerized FastAPI services on Docker, supporting both batch and near-real-time prediction workflows for internal users.
- Set up experiment tracking and model versioning in MLflow across 8 model iterations, making every result reproducible and easy to compare side by side.
- Implemented drift detection and monitoring jobs that catch distribution shifts on input features, reducing undetected performance regressions by 25%.
- Trained and tuned XGBoost, Random Forest, and Logistic Regression models with cross-validation and Bayesian hyperparameter search via Optuna, beating baseline accuracy by 12% on the held-out set.
- Automated data ingestion, validation, and feature engineering pipelines using Python, Pandas, SQL, and AWS S3, ensuring reliable preprocessing for downstream model training and inference workflows.

Machine Learning Engineer

Oct 2020 – Aug 2023

KPMG Global Services

Bengaluru, India

- Built distributed ML pipelines using Python, PySpark, and Databricks to process high-volume transactional records daily for predictive analytics and financial risk applications, reducing pipeline runtime by 30%.
- Trained and tuned supervised and unsupervised models including XGBoost, SVM, K-Means, and PCA with Bayesian hyperparameter optimization, lifting accuracy on client risk-scoring benchmarks by 15%.

- Integrated Kafka and Spark Streaming components to feed live transactional signals into inference pipelines, cutting end-to-end model latency by 20%.
- Deployed batch and near-real-time inference workloads on AWS with containerized Docker workflows, supporting high-throughput ML applications across enterprise client engagements.
- Owned CI/CD for model deployment using Git and Docker, with SHAP-based explainability reports and ROC-AUC and precision-recall dashboards that stakeholders relied on for production sign-off.
- Maintained a centralized feature store on AWS Redshift that kept training and serving features consistent, eliminating feature drift across 5 production models in active client engagements.
- Drove monthly model performance reviews with data science, product, and risk stakeholders, using SHAP-based feature importance and segment-level error analysis to surface drift and bias before it affected business outcomes.
- Scheduled model retraining and validation jobs in Apache Airflow with automated data quality and feature drift checks, ensuring model freshness across 5+ production deployments.

Projects

Smart Quiz Generator – Agentic AI Learning Platform

- Created a multi-agent AI system in LangChain with 5 specialized agents (Transcription, Content Filtering, Concept Extraction, Question Generation, Evaluation) that converts YouTube lectures into concept-grounded MCQ quizzes using Llama 3.3 70B via Groq.
- Combined an LLM-powered Filter Agent with an adaptive learning loop that classifies transcript chunks as educational or filler and tracks per-topic performance across sessions (Mastered, Improving, Needs Work) to generate follow-up quizzes focused on weak areas.
- Exposed the system through a FastAPI backend with endpoints for quiz generation, answer evaluation, and progress tracking, with a Streamlit frontend deployed on Render.

Breast Cancer Survival Prediction System

- Modeled patient survival outcomes using Cox Proportional Hazards and Random Survival Forest (lifelines, scikit-learn, PyTorch), reaching a C-index of 0.81 on held-out cohorts.
- Containerized a FastAPI inference endpoint on Docker for real-time risk scoring, backed by feature engineering pipelines in Pandas and NumPy with missing-value imputation and normalization.
- Surfaced the top 5 clinical biomarkers driving patient risk through SHAP explainability dashboards, giving medical stakeholders actionable model transparency.

Education

Master of Science in Computer Science | California State University, Dominguez Hills, Carson, CA | December 2025

Bachelor of Science in Computer Science | K.S. School of Engineering and Management, Bengaluru, India | August 2020